SOKOINE UNIVERSITY OF AGRICULTURE
School of Engineering and Technology

Course: ABE 721 - Agricultural Systems Modelling and Simulation
Instructor: Kadeghe Fue, PhD, P.Eng(T)

ASSIGNMENT 1
Module 1: Advanced Computer Aided Design and Analysis

Instructions:

- This assignment is designed to improve your foundational understanding before we move into complex numerical integration in Module 2.
- Carefully read the ”Significance” statement for each question to understand the practical engineering application of the theory.
- Show all logical steps in your formulations.

Tier 1: Simple (Conceptual Foundations)

Question 1.1: Geometric Modelling Paradigms

Significance of Question: *Choosing the correct modelling paradigm (Wireframe, Surface, or Solid) dictates the computational efficiency and downstream usability of a CAD file in manufacturing.*

Prompt: Briefly define Constructive Solid Geometry (CSG) and Boundary Representation (B-Rep). Explain why an agricultural engineer designing a complex, load-bearing tractor axle would strictly choose Solid Modelling over Surface Modelling.

Question 1.2: Systems Component Identification

Significance of Question: *Correctly identifying and isolating system components is the absolute mandatory first step; misidentifying an exogenous variable as an endogenous state variable will cause a mathematical model to fail completely.*

Prompt: Consider an automated, thermostatically controlled grain drying silo. Identify and list the following components for this specific system:

a) Two Exogenous Variables (Inputs from the environment).

b) Two State Variables.

c) One physical feedback mechanism.

Tier 2: Intermediate (Application and Logic)

Question 2.1: Bounding the System

Significance of Question: *Defining the system boundary prevents the "infinite complexity" trap. Engineers must mathematically justify what they choose to ignore to make a model solvable.*

Prompt: You are tasked with modelling the fuel consumption of a combine harvester over a 100-hectare field to optimize the machine's forward speed. Draw a basic block diagram defining your System Boundary. Explicitly state two physical factors you have chosen to place *outside* the boundary (ignored in your model) and justify why they are mathematically negligible for this specific objective.

Question 2.2: Finite Element Analysis (FEA) Mechanics

Significance of Question: *FEA software can easily generate colorful, impressive-looking graphics that are entirely wrong. Understanding the underlying matrices prevents the "garbage-in, garbage-out" phenomenon in mechanical design.*

Prompt: In the context of Computer-Aided Engineering (CAE), the fundamental static equation for FEA is $[K]\{U\} = \{F\}$.

a) Define the three matrix/vector terms in this equation.

b) Explain the concept of "meshing" a CAD model. How does the density of the mesh (the number of elements) affect both the computational time and the accuracy of the $\{U\}$ output?

Tier 3: Hard (Synthesis and Formulation)

Question 3.1: Formulating Constraints for Farm Machinery Scheduling

Significance of Question: *Transforming abstract logistical farming problems into strict mathematical inequalities is the foundation of Operations Research. This allows com-*

puter algorithms to find optimal solutions that save millions of shillings in operational costs.

Prompt: A commercial farm must schedule land preparation (ploughing) and planting operations. You have 2 tractors. The farm has 3 distinct fields (F1, F2, F3). Let the binary decision variable $X_{i,j,t} = 1$ if operation i is performed on field j at time t, and 0 otherwise.

Formulate the mathematical constraint equations for the following logistical realities:

a) **Resource Constraint:** You cannot utilize more tractors than you physically own at any given time t.

b) **Precedence Constraint:** For any given field j, the planting operation cannot begin until the ploughing operation has been fully completed.

Note: You do not need to solve the equations; simply write the formal mathematical inequalities.

Question 3.2: Integrating CAD, Sensors, and Systems Logic

Significance of Question: *Modern precision agriculture is not just mechanical drafting; it requires the synthesis of mechanical hardware (CAD), environmental sensors, and dynamic control theory to create intelligent machines.*

Prompt: You are designing a ”Variable-Rate Fertilizer Applicator” implement.

a) Propose a mechanical metering mechanism that could be designed using CAD.

b) Design a closed-loop system architecture (using a block diagram or text description) that utilizes a GPS spatial map (Parameter), a tractor ground-speed radar (Input), and a hopper mass-flow sensor (Feedback) to continuously adjust your mechanical metering mechanism (Process) to achieve a precise target application rate (Output).

Please submit your work to my email kadeghe.fue@sua.ac.tz by one week after completion of Module 1. Do not send this work to kadefue@sua.ac.tz.